

Ch12: Ethical Aspects of Emerging and Converging Technologies

P4: Machine Ethics and (Designing) Moral Machines



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4. Machine Ethics and (Designing) Moral Machines

- Anderson and Anderson (2011) describe machine ethics as an interdisciplinary field of research that is primarily concerned with developing ethics for machines, as opposed to developing ethics for humans who “use machines.”
- In their view, machine ethics is concerned with giving machines ethical principles, or a procedure for discovering ways to resolve ethical dilemmas they may encounter, enabling them to function in an ethically responsible manner through their own decision making.

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Machine Ethics (Continued)

- Wallach and Allen’s main interest in how AMs can be made into moral reasoners is more practical than theoretical in nature.
- Wallach and Allen also believe that the challenge of figuring out how to provide software/hardware agents with moral decision-making capabilities is urgent.
- In fact, they argue that the time to begin work on designing “moral machines” is now!

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Module Objectives

Upon completion of this lesson the students should be able to:

- Discuss **Robotic ethics**
 - ❖ Recall Asimov’s Three Laws of Robotics
 - ❖ Describe and discuss the issues in creation of machines capable of making moral decisions

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Machine Ethics (Continued)

- Wallach and Allen (2009) believe that one way in which the field of machine ethics has expanded upon traditional computer ethics is by asking how computers can be made into “explicit moral reasoners.”
- In their answer to this question, Wallach and Allen first draw an important distinction between “reasoning about ethics” and “ethical decision making.”
- For example, they acknowledge that even if one could build artificial systems capable of reasoning about ethics, it does not necessarily follow that these systems would be genuine “ethical decision makers.”

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Moral Machines

- Can/should we build the kinds of “moral machines” that Wallach and Allen urge us to develop?
- The kind of moral machines that Wallach and Allen have in mind are AMs that are capable of both
 - a) **making moral decisions;**
 - b) **acting in ways that “humans generally consider to be ethically acceptable behavior.”**
- We should note that the idea of designing machines that could behave morally, i.e., with a set of moral rules embedded in them, is not entirely new.

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Designing "Moral Machines"

- In the 1940s, Isaac Asimov anticipated the need for ethical rules that would guide the robots of the future.
- He then formulated his (now classic) Three Laws of Robots:
- 1. A robot may not injure a human being, or through inaction, allow a human being to come to harm.
- 2. A robot must obey orders given it by human beings except where such orders would conflict with the First Law.
- 3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.

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Designing Moral Machines (Continued)

- Numerous critics have questioned whether the three laws articulated by Asimov are adequate to meet the kinds of ethical challenges that current AMs pose.
- But relatively few of these critics have proposed clear and practical guidelines for how to embed machines with ethical instructions that would be generally acceptable to most humans.
- Susan and Michael Anderson (2011) and Wallach and Allen (2009) have each put forth some very thoughtful proposals for how this can be done.

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Designing Moral Machines (Continued)

- S. Anderson argues that it would be prudent for us first to design an artificial system to function as an "ethical advisor" to humans before building a full-fledged moral machine.
- S. Anderson and M. Anderson (2011) have recommended building artificial systems with which humans can have an "ethical dialogue" before we embed machines themselves with ethical reasoning algorithms that they could use in a fully independent manner.

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Designing Moral Machines (Continued)

- The Andersons have developed an "automated dialogue" – i.e., a system involving an ethicist and an artificial system that functions "more or less independently in a *particular domain*."
- They believe that this is an important first step in building moral machines because it enables the artificial system to learn both:
 - a) the "ethically relevant features of the dilemmas it will encounter" (within that domain),
 - b) the appropriate prima facie duties and decision principles it will need to resolve the dilemmas.

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Designing Moral Machines

- Wallach and Allen seem far less concerned with questions about whether AMs can be full moral agents than with questions about how we can design AMs to act in ways that conform to our received notions of morally acceptable behavior.
- S. Anderson (2011) echoes this point when she notes that her primary concern also is with whether machines "can perform morally correct actions and can justify them if asked."

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Designing "Moral Machines" and the Importance of Machine Ethics

- Why is continued work on designing moral machines in particular, an in machine ethics in general, important?
- Moor (2006) proposes three reasons:
 - 1) ethics (itself) is important;
 - 2) future machines will likely have increased autonomy;
 - 3) designing machines to behave ethically will help us better understand ethics.
- Moor's third reason reinforces Wallach and Allen's claim that developments in machine ethics could also help us to better understand our own nature as moral reasoners.

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5. A "Dynamic" Ethical Framework for Guiding Research in New and Emerging Technologies

- Some of the ethical concerns affecting AMs, as well as the other new/emerging technologies that we have examined, directly impact the software engineers/programmers who design the technologies.
- But virtually everyone will be affected by these technologies in the near future.
- So, we all would benefit from clear ethical guidelines that address research/development in new and emerging technologies.

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The Need for Clear Ethical Guidelines for Emerging Technologies (Continued)

- What kind of ethical framework works best for research involving emerging technologies?
- Ray Kurzweil (2005) believes that an ELSI-like model should be developed and used to guide researchers working in one area of emerging technologies – nanotechnology.
- Many consider the ELSI framework to be an ideal model because it is a "proactive" (rather than a reactive) ethics framework.

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The Need for Ethical Guidelines

- Moor also argues that because new and emerging technologies promise "dramatic change," it is no longer satisfactory to do "ethics as usual."
- Instead, he claims that we need to be:
- better informed in our "ethical thinking";
- more proactive in our "ethical action."

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The Need for Ethical Guidelines for Emerging Technologies (Continued)

- As research began on the Human Genome Project (HGP) in the 1990s, researchers developed an ethical framework that came to be known as ELSI (Ethical, Legal, and Social Issues) to anticipate (i.e., in advance) some HGP-related ethical issues that would likely arise.
- Prior to the ELSI Program, ethics was typically "reactive" in the sense that it had followed scientific developments, rather than informing scientific research.
- For example, Moor (2004) notes that in most scientific research areas, ethics has had to play "catch up," because guidelines were developed in response to cases where serious harm had already resulted.

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The Need for Ethical Guidelines (Continued)

- But Moor (2008) is critical of the ELSI model because it employs a scheme that he calls an "ethics-first" framework.
- He believes that this kind of ethical framework has problems because:
 - a) it depends on a "factual determination" of the specific harms and benefits of a technology before an ethical assessment can be done;
 - b) in the case of nanotechnology, it is very difficult to know what the future will be.

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The Need for Ethical Guidelines

- Moor and Weckert (2004) note that if we use an ELSI-like ethics model, it might seem appropriate to put a moratorium on research in a specific new/emerging technology until we get all of the facts.
- But they also believe that while a moratorium would halt technology developments, it will not advance ethics in that area of emerging technologies.

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The Need for Ethical Guidelines

- Moor and Weckert also argue that turning back to a traditional “ethics-last model” is not desirable either.
- They note that once a technology is in place, much unnecessary harm may already have occurred.
- So, for Moor and Weckert, neither an ethics-first (i.e., ELSI-like) nor an ethics-last model is satisfactory for emerging technologies.

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A “Dynamic” Ethical Framework That Continually Needs to be Updated

- Moor and Weckert argue that ethics is something that needs to *be done continually* as:
- a specific (new) technology develops,
- that technology’s potential consequences become better understood.
- The authors also point out that ethics is “dynamic” in that the factual component on which it relies has to be continually updated.

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Moor’s Dynamic Ethical Framework for New/Emerging Technologies

- Building on Moor’s and on Moor and Weckert’s insights regarding ethical challenges posed by new and emerging technologies, we add a fourth component or step to our (threefold) framework developed in Chapter 1, that is:
- Step 4. *Update the ethical analysis* by continuing to:
 - a) differentiate between the factual/descriptive and normative components of the new or emerging technology under consideration;
 - b) revise the policies affecting that technology as necessary, especially as the factual data changes or as information about the potential social impacts becomes clearer.

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Moor’s Dynamic Ethical Framework (Continued)

- As information about plans for the design and development of a (particular) new technology becomes available, we can:
- loop back to Step 1 (of our model in Chap. 1)
- proceed carefully through each subsequent step in the expanded ethical framework.
- This four-step framework can also be applied as new information about existing technologies and their features becomes available.

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